

In the field of Information and Communication Technology (ICT), a linker, also known as a linker program or simply a linker, is a software tool that plays a crucial role in the process of compiling and linking source code written in a programming language into executable programs or libraries. Here's an overview of what a linker does:

- **Compilation:** When you write a program in a high-level programming language like C, C++, or Rust, you typically create one or more source code files (.c, .cpp, .rs, etc.). These source code files contain human-readable code written by the programmer. The first step in turning this code into a runnable program is compilation.
- **Compilation Process:** The compilation process involves a compiler, which translates the human-readable source code into machine code or intermediate code. This results in one or more object files.
- **Linking:** This is where the linker comes into play. The linker takes one or more object files, along with any necessary system libraries or user-defined libraries, and combines them to produce an executable program or a shared library (DLL or .so file). During this process, the linker resolves references and symbols, ensuring that functions and variables used in one object file can properly refer to definitions in other object files or libraries.

IN BRIEF:

A linker in Information and Communication Technology (ICT) is a software tool that connects and combines multiple object files, generated from source code, into a complete executable program or library. It resolves references between different parts of the program, assigns memory addresses, adjusts references, and eliminates dead code to produce a functional and runnable software application. Linkers play a crucial role in the software development process, ensuring that high-level programming code can be transformed into executable programs that can run on a computer or device.